

Last time $f: X \rightarrow Y$, $x_0 \in X$

f continuous at $x_0 \iff \lim_{x \rightarrow x_0} f(x) = f(x_0)$

$\iff \forall \varepsilon > 0, \exists \delta > 0$ s.t. $0 < d_X(x, x_0) < \delta$ implies $d_Y(f(x), f(x_0)) < \varepsilon$

$\iff \forall$ sequence $\{x_n\}_{n \geq 1}$ in X converging to x_0 ,
the sequence $\{f(x_n)\}_{n \geq 1}$ in Y converges to $f(x_0)$

Cor Let X be a metric space. If $f, g: X \rightarrow \mathbb{R}$ are continuous at x_0 , then:

- ① $k \cdot f$ is continuous at x_0 , $\forall k \in \mathbb{R}$.
- ② $f + g$ is continuous at x_0
- ③ $f \cdot g$ is continuous at x_0
- ④ $\frac{f}{g}$ is continuous at x_0 , if also $g(x_0) \neq 0$.

Pf: Fix a sequence $x_n \rightarrow x_0$ in X as $n \rightarrow \infty$. Then $\{f(x_n)\}_{n \geq 1}$ and $\{g(x_n)\}_{n \geq 1}$ are sequences of real numbers that converge to $f(x_0)$ and $g(x_0)$, respectively. Now apply the limit laws (Lectures 10-11). $\square$

Cor If $p: \mathbb{R} \rightarrow \mathbb{R}$ is a polynomial:

$$p(x) = c_0 + c_1 x + \dots + c_N x^N, \quad N \in \mathbb{N} \cup \{0\}, \quad c_0, \dots, c_N \in \mathbb{R}$$

then p is continuous at any $x_0 \in \mathbb{R}$.

Pf: Fix a convergent sequence of real numbers, and apply the limit laws. $\square$

Def Let X, Y be metric spaces. We say $f: X \rightarrow Y$ is continuous if it is continuous at $x_0 \forall x_0 \in X$.

Prop Let X, Y be metric spaces and $f: X \rightarrow Y$.

Then: f is continuous $\Leftrightarrow \forall$ open set $U \subseteq Y$, $f^{-1}(U)$ is open in X .

Pf: $\Rightarrow$: Suppose f is continuous. Fix U open in Y and $x_0 \in f^{-1}(U)$. As $f(x_0) \in U$, then $\exists \varepsilon > 0$ s.t. $B_\varepsilon^Y(f(x_0)) \subseteq U$. As $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, then $\exists \delta > 0$ s.t.

$$x \in X, \quad 0 < d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

$$\text{Don't need: } x = x_0 \Rightarrow d_Y(f(x_0), f(x_0)) = 0 < \varepsilon$$

Then $B_\delta^X(x_0) \subseteq f^{-1}(U)$, since:

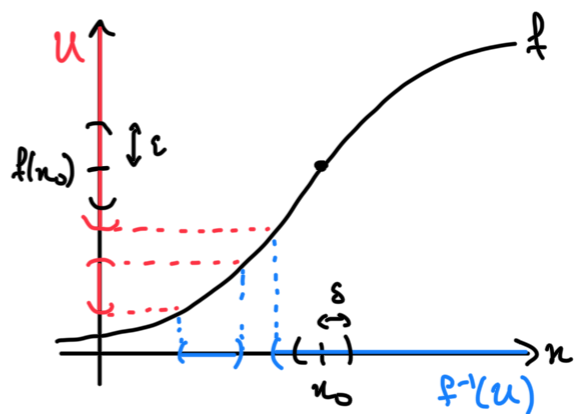
$$x \in B_\delta^X(x_0) \Rightarrow d_X(x, x_0) < \delta$$

$$\Rightarrow d_Y(f(x), f(x_0)) < \varepsilon$$

$$\Rightarrow f(x) \in B_\varepsilon^Y(f(x_0)) \subseteq U$$

$$\Rightarrow x \in f^{-1}(U).$$

So $f^{-1}(U)$ is open.



$\Leftarrow$: Suppose $\forall U \subseteq Y$ open, $f^{-1}(U)$ is open in X .

Fix $x_0 \in X$. Want: f is continuous at x_0 . Fix $\varepsilon > 0$.

Set $U = B_\varepsilon^Y(f(x_0))$. As U is open in Y , we know $f^{-1}(U)$ is open in X . Note that $x_0 \in f^{-1}(U)$, since $f(x_0) \in B_\varepsilon^Y(f(x_0)) = U$. So $\exists \delta > 0$ s.t. $B_\delta^X(x_0) \subseteq f^{-1}(U)$.

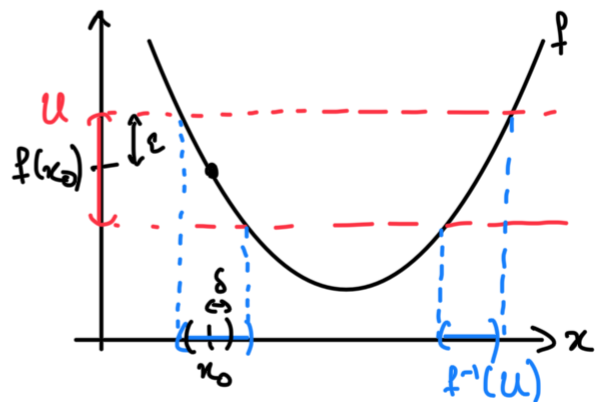
Then:

$$0 < d_X(x, x_0) < \delta$$

$$\Rightarrow x \in B_\delta^X(x_0) \subseteq f^{-1}(U)$$

$$\Rightarrow f(x) \in U = B_\varepsilon^Y(f(x_0))$$

$$\Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$



So $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. $\square$

Cor Let X, Y be metric spaces and $f: X \rightarrow Y$.

Then: f is continuous $\Leftrightarrow \forall$ closed set $F \subseteq Y$, $f^{-1}(F)$ is closed in X .

Pf: Apply the previous proposition to $U = F^c$.

Note that $f^{-1}(F^c) = (f^{-1}(F))^c$:

$$x \in f^{-1}(F^c) \Leftrightarrow f(x) \notin F \Leftrightarrow x \notin f^{-1}(F). \quad \square$$